

Elijah Morales

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SUMMARY

Applied Mathematics M.S. student at Columbia and NSF REU alumnus with experience in stochastic modeling, scientific computing, structured data verification, and workflow analytics. Built and analyzed population-dynamics models using Python/MATLAB, ODE/PDE formulations, stochastic simulation, and evolutionary game theory. Currently developing internal tools, dashboards, and human-in-the-loop automation for background-screening operations. Seeking quantitative analyst, quantitative data science, risk analytics, and applied research roles.

EDUCATION

- **Columbia University** Jan 2026 - May 2028 expected
New York, NY
M.S. in Applied Mathematics
 - GPA: 4.17 (Columbia scale)
 - Completed: APMA E4008 Advanced Linear Algebra; APMA E4300 Introduction to Numerical Methods.
Current/planned focus: probability/statistics, stochastic models, optimization, machine learning, financial engineering, PDEs, applied stochastic analysis.
- **University of South Florida, Judy Genshaft Honors College** Aug 2022 - May 2025
Tampa, FL
B.A. in Pure Mathematics & Psychology
 - GPA: 3.81/4.00
 - Graduated in three years; Magna Cum Laude; Honors College Graduate; Dean's List all semesters.
 - Coursework: Intermediate Analysis I-II, Abstract Algebra I-II, Topology, PDEs, Differential Equations, Linear Algebra, Vector Calculus, Symbolic Computation, Programming Concepts (C++), Calculus I-III.

EXPERIENCE

- **SRA Screening**  Oct 2025 - Present
West Palm Beach, FL
Research Analyst - Internal Tools, Automation & Analytics
 - Perform structured verifications across employment, education, legal, licensing, financial, and reference records while maintaining compliance standards and audit-trail integrity.
 - Investigate discrepancies using SQL-based screening platforms, Excel, public records, registry databases, and internal knowledge resources.
 - Designed and am building an internal operations portal centralizing department knowledge, dashboards, trackers, workflow tools, SOPs, and reference materials.
 - Developed workflow improvements for case tracking, vendor dispatch logic, turnaround monitoring, and researcher handoff support.
 - Prototyping human-in-the-loop automation to reduce repetitive pre-research tasks, identify missing information, and standardize case preparation before researcher review.
- **Arizona State University - NSF QRLSSP REU**  May 2024 - Jul 2024
Tempe, AZ
Quantitative Research Assistant
 - Selected for NSF-funded Quantitative Research in the Life and Social Sciences Program focused on stochastic systems and structured population models.
 - Modeled dispersal evolution in *Tribolium* metapopulations using discrete-time LPA dynamics, ODE/PDE models, stochastic simulation, and evolutionary game theory.
 - Developed and analyzed 4-patch and 5-patch metapopulation simulations with adult dispersal, dispersal cost, local extinction events, and mutant/resident strategies.
 - Derived and interpreted R_0 -style viability thresholds and ESS/invasion criteria; visualized stochastic extinction scenarios and polymorphic/monomorphic outcomes.
 - Co-authored a 67-page technical report/preprint and presented findings through a research poster and oral presentation.

TECHNICAL SKILLS

- **Languages & Tools:** Python, SQL, MATLAB, LaTeX, Git/GitHub, Excel, PowerQuery, C++ coursework, Maple
- **Scientific Computing:** NumPy, SciPy, matplotlib, numerical methods, Monte Carlo simulation, ODE/PDE modeling, least squares, optimization, stochastic simulation
- **Quant/Data Concepts:** Stochastic modeling, Markov-chain concepts, population dynamics, evolutionary game theory, data verification, workflow analytics, dashboarding
- **Certifications:** SQL Certificate of Competency - LearnSQL.com, Jun 2025; CITI Human Subjects Research, Jan 2023

RESEARCH

- **K. Angell, E. Morales, B. Wiltshire, J. Rodriguez Rincon, L. Rodriguez Rodriguez, J.D. Nagy.** – *Dispersal evolution in Tribolium metapopulations: a game theory approach*. QRLSSP Technical Report / Preprint, Arizona State University, 2024.